

# Appendix J - Full Memory Map

The fixed low regions of the Intuition Engine 64-bit physical bus, in address order, with their size and purpose. IE64 can address beyond this low window when RAM is present. IE32, M68K, and x86 see the low 4 GB window directly; the 6502 and Z80 see windowed views into it (Chapters 27 and 28).

## J.1 Low RAM and stacks

| Range             | Size    | Purpose                                                |
|-------------------|---------|--------------------------------------------------------|
| \$000000-\$0003FF | 1 KB    | M68K vector table; x86 IVT; trap vector base for IE64. |
| \$000400-\$022FFF | ~140 KB | BASIC ROM image, system call shim, monitor stub.       |
| \$023000-\$04FFFF | 180 KB  | BASIC program text (BASIC_PROG_START-).                |
| \$050000-\$057FFF | 32 KB   | BASIC simple variables.                                |
| \$058000-\$05FFFF | 32 KB   | BASIC string variables.                                |
| \$060000-\$08BFFF | 176 KB  | BASIC arrays.                                          |
| \$08C000-\$08FFFF | 16 KB   | BASIC string temporaries.                              |
| \$090000-\$096FFF | 28 KB   | BASIC GOSUB / FOR stack.                               |
| \$097000-\$09EFFF | 32 KB   | Free RAM (general user).                               |
| \$09F000-\$09FFFF | 4 KB    | IE32 stack (STACK_START).                              |

## J.2 PC-compatible VRAM apertures

| Range             | Size  | Purpose                                                           |
|-------------------|-------|-------------------------------------------------------------------|
| \$0A0000-\$0AFFFF | 64 KB | VGA graphics window (mode 13h, mode 12h linear).                  |
| \$0B0000-\$0B7FFF | 32 KB | Reserved (PC-compatible monochrome text window; not implemented). |
| \$0B8000-\$0BFFFF | 32 KB | VGA text buffer (\$B8000).                                        |

## J.3 General RAM gap

| Range             | Size  | Purpose                  |
|-------------------|-------|--------------------------|
| \$0C0000-\$0CFFFF | 64 KB | Free RAM (general user). |
| \$0D0000-\$0DFFFF | 64 KB | Voodoo texture RAM.      |
| \$0E0000-\$0EFFFF | 64 KB | Free RAM (general user). |

## J.4 The MMIO region (\$F0000-\$FFFFF)

| Range           | Size   | Device                                   |
|-----------------|--------|------------------------------------------|
| \$F0000-\$F049B | 1180 B | Video chip + palette + extended blitter. |
| \$F0700-\$F07FF | 256 B  | Terminal / serial / input.               |
| \$F0800-\$F0B7F | 896 B  | SoundChip channel registers.             |
| \$F0B80-\$F0B91 | 18 B   | AHX player.                              |
| \$F0BA0-\$F0BBF | 32 B   | MIDI/MUS player.                         |
| \$F0BC0-\$F0BD7 | 24 B   | MOD player.                              |
| \$F0BD8-\$F0BF3 | 28 B   | WAV player.                              |
| \$F0C00-\$F0C20 | 33 B   | PSG / AY.                                |
| \$F0C30-\$F0C3F | 16 B   | SN76489.                                 |
| \$F0C40-\$F0CFF | 192 B  | SoundChip flex channels 4-6.             |
| \$F0D00-\$F0D20 | 33 B   | POKEY.                                   |
| \$F0D40-\$F0DFF | 192 B  | SoundChip flex channels 7-9.             |
| \$F0E00-\$F0E1C | 29 B   | SID primary registers.                   |
| \$F0E20-\$F0E2D | 14 B   | SID player block.                        |
| \$F0E30-\$F0E4C | 29 B   | SID2 registers.                          |
| \$F0E50-\$F0E6C | 29 B   | SID3 registers.                          |
| \$F0E80-\$F0EFF | 128 B  | SFX triggers.                            |
| \$F0F00-\$F0F6B | 108 B  | TED audio + video.                       |
| \$F1000-\$F13FF | 1 KB   | VGA registers.                           |
| \$F1400-\$F140F | 16 B   | HOST appliance block.                    |
| \$F2000-\$F2017 | 24 B   | ULA registers.                           |
| \$F2100-\$F213F | 64 B   | ANTIC.                                   |
| \$F2140-\$F21FB | 188 B  | GTIA.                                    |
| \$F2200-\$F221F | 32 B   | File I/O.                                |
| \$F2260-\$F22AF | 80 B   | Amiga Paula DMA.                         |
| \$F2300-\$F231F | 32 B   | Media loader.                            |
| \$F2320-\$F233F | 32 B   | RUN loader block.                        |
| \$F2340-\$F238F | 80 B   | Coprocessor.                             |
| \$F2390-\$F23AF | 32 B   | Clipboard bridge.                        |
| \$F23B0-\$F23BF | 16 B   | Coprocessor extended monitor.            |
| \$F23C0-\$F23DF | 32 B   | IRQ diagnostics.                         |
| \$F23E0-\$F23FF | 32 B   | Bootstrap loader.                        |

| Range           | Size   | Device                  |
|-----------------|--------|-------------------------|
| \$F2400-\$F24FF | 256 B  | SysInfo (RAM-size ABI). |
| \$F8000-\$F87FF | 2 KB   | Voodoo 3D registers.    |
| \$FA000-\$FBAFF | 6912 B | ULA VRAM aperture.      |

## J.5 Main video RAM

| Range              | Size | Purpose                                                                                               |
|--------------------|------|-------------------------------------------------------------------------------------------------------|
| \$100000-\$5FFFFFF | 5 MB | Main VRAM aperture for VideoChip framebuffers; large modes may point VIDEO_FB_BASE into ordinary RAM. |

## J.6 Low-window RAM and reserved aliases

| Range                 | Size    | Purpose                                                       |
|-----------------------|---------|---------------------------------------------------------------|
| \$00600000-\$FFFEFFFF | Dynamic | Extended RAM when backed by the current guest-RAM allocation. |
| \$FFFF0000-\$FFFFFFFF | 64 KB   | Sign-extended alias of \$00000000-\$0000FFFF.                 |

Guest RAM can extend beyond these fixed low ranges. Use the SysInfo registers in Chapter 24 to discover total and active visible RAM. IE64 can address backed RAM above \$FFFFFFFF; the compatibility CPUs remain inside the low window or their documented profile caps.

## J.7 The 6502 / Z80 views

The 6502 sees a 16-bit address space \$0000-\$FFFF. Its adapter routes the high page-window region as follows:

| 6502 range    | Maps to low bus address                                                                      |
|---------------|----------------------------------------------------------------------------------------------|
| \$0000-\$00FF | zero page in main RAM.                                                                       |
| \$0100-\$01FF | 6502 stack page in main RAM.                                                                 |
| \$0200-\$CFFF | main RAM.                                                                                    |
| \$D200-\$D20A | POKEY registers.                                                                             |
| \$D400-\$D40F | PSG registers.                                                                               |
| \$D500-\$D55F | SID registers.                                                                               |
| \$D600-\$D605 | TED audio registers.                                                                         |
| \$D620-\$D632 | TED video registers.                                                                         |
| \$D700-\$D70D | VGA registers.                                                                               |
| \$D800-\$D817 | ULA registers, including the paged VRAM data port.                                           |
| \$E000-\$EFFF | bank-selectable window into main RAM.                                                        |
| \$F000-\$FFF9 | MMIO mirror of \$F0000-\$F0FF9, with \$F700-\$F705 and \$F7F0 intercepted as bank registers. |
| \$FFFA-\$FFFF | NMI / reset / IRQ vectors.                                                                   |

The Z80 sees the same shape with port I/O substituted for the chip-aperture region; the bank-window scheme is identical.

## J.8 Notes

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The map above is the compact engineering view of the low and device regions that have fixed meanings inside the 64-bit physical bus. Most user programs use a small sub-set of it - the BASIC variable area, the I/O region, and the main video RAM - and never touch the rest. Programs that compete for memory (a large in-memory dataset, a custom blitter routine, a coprocessor worker that wants its own heap) should carve their storage out of the "Free RAM" gaps in J.1, J.3, and J.6, or from boot-sized extended RAM reported by SysInfo, rather than colliding with the BASIC variable region.